Differential Geometry
Solved Past Paper

Unit Speed Curves · Frenet–Serret Apparatus · Non-Unit Speed Curves

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Total Marks:	50 (Attempt any **Four**, two from each section)
Time:	2 Hours

Section A – Unit Speed Curves

Q.No. 1
Regular Curves and Re-parametrization

(a) Define a regular curve and a re-parametrization of a curve.

Solution

A curve $\alpha : (a, b) \to \mathbb{R}^3$ is **regular** if $\alpha'(t) \neq \mathbf{0}$ for all $t \in (a, b)$.
A **re-parametrization** is a diffeomorphism $g : (c, d) \to (a, b)$ (smooth bijection with $g' \neq 0$). The curve $\beta = \alpha \circ g$ is then a re-parametrization of α.

(b) Let $\beta = \alpha \circ g$, $t_0 = g(s_0)$. Show that the unit tangent S of β at s_0 satisfies $S = \pm T$, where T is the unit tangent of α at t_0.

Solution

Step 1. Apply the chain rule to $\beta = \alpha \circ g$:

$$\beta'(s_0) = \alpha'(g(s_0)) \cdot g'(s_0) = \alpha'(t_0) \cdot g'(s_0).$$

Step 2. Find the unit tangent of β:

$$S = \frac{\beta'(s_0)}{|\beta'(s_0)|} = \frac{\alpha'(t_0) \cdot g'(s_0)}{|\alpha'(t_0)| \cdot |g'(s_0)|} = \frac{g'(s_0)}{|g'(s_0)|} \cdot T = \pm T.$$

Conclusion: $S = +T$ if $g'(s_0) > 0$ (same direction), $S = -T$ if $g'(s_0) < 0$ (opposite direction). □

(c) Show that α and $\beta = \alpha \circ g$ have the same arc length.

Solution

Step 1. Write the arc length of α:

$$L(\alpha) = \int_a^b |\alpha'(t)|\, dt.$$

Step 2. Substitute $t = g(s)$, so $dt = g'(s)\, ds$. When t goes from a to b, s goes from c to d:

$$L(\alpha) = \int_c^d |\alpha'(g(s))|\, |g'(s)|\, ds = \int_c^d |\alpha'(g(s)) \cdot g'(s)|\, ds = \int_c^d |\beta'(s)|\, ds = L(\beta).$$

Conclusion: $L(\alpha) = L(\beta)$. Arc length does not depend on parametrization. $\square$

Q.No. 2

Torsion and Curvature of a Plane Curve

(a) Let $\alpha(s)$ be a C^4 unit-speed curve in the (x, y)-plane with $\kappa \neq 0$. Prove $\tau \equiv 0$.

Solution

Step 1. Write $\alpha(s) = (x(s), y(s), 0)$. Compute the Frenet vectors:

$$T = \alpha' = (x', y', 0), \qquad \kappa N = \alpha'' = (x'', y'', 0).$$

Step 2. Compute the binormal $B = T \times N$:

$$B = \frac{1}{\kappa}\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x' & y' & 0 \\ x'' & y'' & 0 \end{vmatrix} = \frac{1}{\kappa}(0,\ 0,\ x'y'' - y'x'') \cdot \hat{k} = \pm(0, 0, 1) = \pm\mathbf{k}.$$

Step 3. Since the curve lies in the $z = 0$ plane, $B = \pm\mathbf{k}$ is a *constant* vector. So:

$$B'(s) = \mathbf{0} \implies -\tau N = \mathbf{0} \implies \tau = 0 \quad (\text{since } \kappa \neq 0 \Rightarrow N \neq \mathbf{0}). \quad \square$$

(b) Let $\alpha(s) = (x(s), y(s), 0)$ be unit-speed. Prove $\kappa = |x'y'' - x''y'|$.

Solution

Step 1. Since α is unit-speed: $|\alpha'| = 1$, so

$$x'^2 + y'^2 = 1.$$

Step 2. Differentiate the above:

$$x'x'' + y'y'' = 0.$$

Step 3. Apply the Lagrange identity $|a \times b|^2 = |a|^2|b|^2 - (a \cdot b)^2$ with $a = (x', y')$, $b = (x'', y'')$:

$$|x'y'' - x''y'|^2 + (x'x'' + y'y'')^2 = (x'^2 + y'^2)(x''^2 + y''^2).$$

Step 4. Substitute Steps 1 and 2:

$$|x'y'' - x''y'|^2 + 0 = 1 \cdot (x''^2 + y''^2) = |\alpha''|^2 = \kappa^2.$$

Therefore $\kappa = |x'y'' - x''y'|$. $\square$

Q.No. 3

Osculating Planes at $s=0$ [Equation: $(\mathbf{r}-\alpha(0))\cdot B(0)=0$]

(a) $\alpha(s)=\left(\dfrac{5}{13}\cos s,\ \dfrac{8}{13}-\sin s,\ -\dfrac{12}{13}\cos s\right)$

Solution

$\alpha'(s)=(-\frac{5}{13}\sin s,-\cos s,\frac{12}{13}\sin s)$, $\quad |\alpha'|^2=\frac{25}{169}\sin^2 s+\cos^2 s+\frac{144}{169}\sin^2 s=1.$ ✓
At $s=0$:

$$\alpha(0)=\left(\tfrac{5}{13},\tfrac{8}{13},-\tfrac{12}{13}\right),\quad T=(0,-1,0),\quad \alpha''(0)=\left(-\tfrac{5}{13},0,\tfrac{12}{13}\right)\Rightarrow\kappa=1,\ N=\left(-\tfrac{5}{13},0,\tfrac{12}{13}\right).$$

$$B=T\times N=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0&-1&0\\-\frac{5}{13}&0&\frac{12}{13}\end{vmatrix}=\left(-\tfrac{12}{13},\ 0,\ -\tfrac{5}{13}\right).$$

Osculating plane:

$$-\tfrac{12}{13}(x-\tfrac{5}{13})-\tfrac{5}{13}(z+\tfrac{12}{13})=0\implies\boxed{12x+5z=0}.$$

(b) $\alpha(s)=\dfrac{1}{\sqrt{5}}\left(\sqrt{1+s^2},\ 2s,\ \ln(s+\sqrt{1+s^2})\right)$

Solution

$\alpha'(s)=\dfrac{1}{\sqrt{5}}\left(\dfrac{s}{\sqrt{1+s^2}},2,\dfrac{1}{\sqrt{1+s^2}}\right)$, $\quad |\alpha'|^2=\dfrac{1}{5}\left(\dfrac{s^2+1}{1+s^2}+4\right)=1.$ ✓
At $s=0$:

$$\alpha(0)=\left(\tfrac{1}{\sqrt5},0,0\right),\quad T=\left(0,\tfrac{2}{\sqrt5},\tfrac{1}{\sqrt5}\right),\quad\alpha''(0)=\left(\tfrac{1}{\sqrt5},0,0\right)\Rightarrow\kappa=\tfrac{1}{\sqrt5},\ N=(1,0,0).$$

$$B=T\times N=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0&\frac{2}{\sqrt5}&\frac{1}{\sqrt5}\\1&0&0\end{vmatrix}=\left(0,\ \tfrac{1}{\sqrt5},\ -\tfrac{2}{\sqrt5}\right).$$

Osculating plane:

$$\tfrac{y}{\sqrt5}-\tfrac{2z}{\sqrt5}=0\implies\boxed{y=2z}.$$

Section B – Non-Unit Speed Curves

Q.No. 4
Frenet Formulas for a Regular (Non-Unit Speed) Curve $\beta(t)$ in $\mathbb{R}^3$

Let $\dot\beta, \ddot\beta, \dddot\beta$ denote derivatives w.r.t. t, and $\dot s = |\dot\beta|$. Key expansion: $\dot\beta = \dot s T$, $\ddot\beta = \ddot s T + \dot s^2 \kappa N$, giving $\dot\beta \times \ddot\beta = \dot s^3 \kappa B$.

(i) Show $B = \dfrac{\dot\beta \times \ddot\beta}{|\dot\beta \times \ddot\beta|}$.

Solution

Step 1. Express $\dot\beta$ and $\ddot\beta$ in the Frenet frame. Since $\dot s = |\dot\beta|$:

$$\dot\beta = \dot s\, T, \qquad \ddot\beta = \ddot s\, T + \dot s^2 \kappa N.$$

Step 2. Take the cross product (using $T \times T = 0$ and $T \times N = B$):

$$\dot\beta \times \ddot\beta = (\dot s\, T) \times (\ddot s\, T + \dot s^2 \kappa N) = \dot s \ddot s (T \times T) + \dot s^3 \kappa (T \times N) = \dot s^3 \kappa\, B.$$

Step 3. Divide by its magnitude ($|B| = 1$, so $|\dot\beta \times \ddot\beta| = \dot s^3 \kappa$):

$$\frac{\dot\beta \times \ddot\beta}{|\dot\beta \times \ddot\beta|} = \frac{\dot s^3 \kappa\, B}{\dot s^3 \kappa} = B. \quad \square$$

(ii) Show $\kappa = \dfrac{|\dot\beta \times \ddot\beta|}{|\dot\beta|^3}$.

Solution

Step 1. From part (i): $|\dot\beta \times \ddot\beta| = \dot s^3 \kappa$.
Step 2. Since $\dot s = |\dot\beta|$, we have $\dot s^3 = |\dot\beta|^3$. So:

$$|\dot\beta \times \ddot\beta| = |\dot\beta|^3 \kappa \implies \kappa = \frac{|\dot\beta \times \ddot\beta|}{|\dot\beta|^3}. \quad \square$$

(iii) Show $\tau = \dfrac{[\dot\beta, \ddot\beta, \dddot\beta]}{|\dot\beta \times \ddot\beta|^2}$.

Solution

Step 1. Differentiate $\ddot\beta = \ddot s\,T + \dot s^2\kappa N$ using $\dot N = \dot s(-\kappa T + \tau B)$:

$$\dddot\beta = (\dddot s - \dot s^3\kappa^2)T + (\text{some coefficient})N + \dot s^3\kappa\tau\,B.$$

The key point: the B-component of $\dddot\beta$ is $\dot s^3\kappa\tau$.

Step 2. Write all three in the $\{T, N, B\}$ frame and take the triple product:

$$[\dot\beta, \ddot\beta, \dddot\beta] = \det\begin{pmatrix} \dot s & 0 & 0 \\ \ddot s & \dot s^2\kappa & 0 \\ & * & \dot s^3\kappa\tau \end{pmatrix} = \dot s\cdot\dot s^2\kappa\cdot\dot s^3\kappa\tau = \dot s^6\kappa^2\tau.$$

Step 3. Since $|\dot\beta\times\ddot\beta|^2 = \dot s^6\kappa^2$:

$$\tau = \frac{[\dot\beta, \ddot\beta, \dddot\beta]}{|\dot\beta\times\ddot\beta|^2}. \quad \square$$

Q.No. 5
Frenet–Serret Apparatus and Plane Curves

(a) Compute the Frenet–Serret apparatus for $\beta(t) = (t - \cos t,\ \sin t,\ t)$.

Solution

Derivatives: $\dot\beta = (1+\sin t, \cos t, 1)$, $\ddot\beta = (\cos t, -\sin t, 0)$, $\dddot\beta = (-\sin t, -\cos t, 0)$.

Speed: $|\dot\beta|^2 = (1+\sin t)^2 + \cos^2 t + 1 = 3 + 2\sin t$.

Unit Tangent:

$$T = \frac{(1+\sin t,\ \cos t,\ 1)}{\sqrt{3+2\sin t}}.$$

Cross product:

$$\dot\beta\times\ddot\beta = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1+\sin t & \cos t & 1 \\ \cos t & -\sin t & 0 \end{vmatrix} = (\sin t,\ \cos t,\ -(1+\sin t)).$$

$$|\dot\beta\times\ddot\beta|^2 = \sin^2 t + \cos^2 t + (1+\sin t)^2 = 1 + (1+\sin t)^2.$$

Binormal:

$$B = \frac{(\sin t,\ \cos t,\ -(1+\sin t))}{\sqrt{1+(1+\sin t)^2}}.$$

Curvature:

$$\kappa = \frac{\sqrt{1+(1+\sin t)^2}}{(3+2\sin t)^{3/2}}.$$

Triple product for torsion:

$$\ddot{\beta} \times \dddot{\beta} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos t & -\sin t & 0 \\ -\sin t & -\cos t & 0 \end{vmatrix} = (0,0,-1), \quad [\dot{\beta}, \ddot{\beta}, \dddot{\beta}] = \dot{\beta} \cdot (0,0,-1) = -1.$$

Torsion:

$$\tau = \frac{-1}{1 + (1+\sin t)^2}.$$

Principal Normal: $N = B \times T$.

(b) Prove β is a plane curve $\iff [\dot{\beta}, \ddot{\beta}, \dddot{\beta}] \equiv 0$ (given $\kappa \neq 0$).

Solution

Part 1 (⇒): Plane curve $\implies$ triple product $= 0$.
Step 1. Suppose β lies in a fixed plane with unit normal $\mathbf{n}$.
Then T and N always stay in that plane, so $B = \pm\mathbf{n} =$ constant.
Step 2. Since B is constant: $B' = \mathbf{0}$. From the Frenet equation $B' = -\tau N$:

$$-\tau N = \mathbf{0} \implies \tau = 0.$$

Step 3. From part (iii) of Q4: $[\dot{\beta}, \ddot{\beta}, \dddot{\beta}] = \dot{s}^6\kappa^2\tau = \dot{s}^6\kappa^2 \cdot 0 = 0.$ ✓

Part 2 (⇐): Triple product $= 0 \implies$ plane curve.
Step 1. If $[\dot{\beta}, \ddot{\beta}, \dddot{\beta}] = 0$, then $\dot{s}^6\kappa^2\tau = 0$.
Since $\dot{s} \neq 0$ (regular curve) and $\kappa \neq 0$ (given), we must have $\tau = 0$.
Step 2. $\tau = 0 \implies B' = -\tau N = \mathbf{0}$, so $B = \mathbf{c}$ (a constant unit vector).
Step 3. Fix any point $\beta(t_0)$. Differentiate $(\beta(t) - \beta(t_0)) \cdot \mathbf{c}$:

$$\frac{d}{dt}\big[(\beta - \beta(t_0)) \cdot \mathbf{c}\big] = \dot{\beta} \cdot \mathbf{c} = \dot{s}\, T \cdot B = 0 \quad (T \perp B \text{ always}).$$

Step 4. So $(\beta(t) - \beta(t_0)) \cdot \mathbf{c} = \text{const} = 0$. This means β lies in the plane:

$$\{p \in \mathbb{R}^3 \ : \ (p - \beta(t_0)) \cdot \mathbf{c} = 0\}. \quad \square$$